Section-15

EARTH WIRE

TECHNICAL SPECIFICATION FOR 7/3.15mm HTGS WIRE

1.0 SCOPE :- Refer Clause-1 of Section-I, General

1.1 This specification provides for manufacture, testing before dispatch, supply and delivery of 7/3.15 mm High Tensile Galvanized stranded steel wire.

1.2 The 7/3.15 mm High Tensile Galvanized stranded steel wire is to be used as earthwire on single, double and Multi circuit 220kV and 132kV transmission lines as well as stay wire on other lines.

1.3 STANDARDS :

The galvanised stranded steel wire shall conform to the following Indian Standards, which shall mean latest revisions, amendments/changes adopted and published unless otherwise specified herein before.

Sl. No.	IS-No.	Title
1.	2141 - 1992	Specification for galvanised stay strand.
2.	209 - 1992	Zinc Ingot specification.
3.	2633 - 1986	Methods for Testing Uniformity of coating on zinc coated Articles.
4.	4826 - 1979	Hot dipped galvanised coatings on round steel wires.
5.	2629 - 1966	
6.	1755 - 1983	Method for wrapping test for metallic wire.
7.	398 (Part-2) – 1976	Specification for Aluminium conductors for overhead Transmission purposes.
8.	6745 - 1972	Methods for determination of mass of zinc coating on zinc coated iron and steel articles.
9.	1778 - 1980	Dimensional drawings of the drum.

2.0 **CONFLICT OF STANDARDS**

Equipment conforming to other internationally accepted standards, which ensure equal or higher quality than the standards mentioned above, would also be acceptable. In case the Bidders who wish to offer material conforming to the other standards, salient points of difference between the standards adopted and the specific standards shall be clearly brought out in relevant schedule. Four copies of such standards with authentic English Translations shall be furnished along with the offer. In case of conflict the order of precedence shall be (i) IS (ii) IEC (iii) Other standards. In case of any

difference between provisions of these standards and provisions of this specification, the provisions contained in this specification shall prevail.

3.0 CLIMATIC CONDITIONS : Refer Clause-2 of Section-I, General

4.0 PRINCIPAL PARAMETERS: The High Tensile galvanised stranded steel wire shall be used as earth wire on single, double & Multi circuit towers for 132kV & 220kV lines in A.P.System.

5.0 THE GENERAL TECHNICAL REQUIREMENTS: The general technical requirements of High Tensile Heavily coated stranded steel wires are tabulated below :

a)	Stranding and wire diameter	7/3.15 mm.
b)	Number of strands:	
	Inner layer	1
	Outer layer	6
c)	Overall diameter	9.45 mm.
d)	Total sectional area	54.57 sq.mm.
e)	Approximate weight	428 Kg./Km.
f)	Minimum breaking load of single wire before stranding	8.57 kN
g)	Minimum breaking load of Earth wire	56.98 kN
h)	Direction of lay of outer layer	Right hand
i)	Tensile grade of wire	Grade-3
j)	Minimum tensile strength	1100 N/Sq.mm.
k)	Minimum percentage of elongation	4
l)	Minimum mass of zinc coating (Heavily coated wire) g/Sq.m.	275
m)	Testing of Uniformity of zinc coating	4 dips of 1 minute duration each
n)	Diameter of individual steel Strand (Std. / Max. / Min.)	3.15/3.21/3.12
o)	Lay length (std/max/min) mm	160 / 175 / 152
p)	Final modulus of elasticity	---- 19330 Kg/Sq.mm ----
q)	Coefficient of linear expansion	11.50 x 10^{-6} per Deg.C

6.0 MATERIALS :

6.1 STEEL :

The wire shall be drawn from steel made by the open hearth basic Oxygen or electic furnace process and of such quality that when drawn to the size of wire specified and coated with zinc, the finished strand and the individual wires shall be of uniform quality and properties and as specified in IS:2141-1979 or its latest revision. The steel wire strands shall be drawn from the steel rods containing the following chemical composition. The wires shall not contain sulphur, phosphorous exceeding 0.060 percent each.

Element	**% Composition**
Carbon	Not more than 0.55
Manganese	0.4 to 0.9
Phosphorous	Not more than 0.04

Sulphur	Not more than 0.04
Silicon	0.15 to 0.35

6.2 ZINC

The zinc used for galvanising shall be Electrolytic High Grade Zinc of 99.95% purity. It shall conform to and satisfy all the requirements of IS:209-1979 or its latest revision. The zinc coating shall confirm to "Heavily Coated" as laid down in IS-4826 and galvanised after drawing without annealing.

7.0 WORKMANSHIP

7.1 All steel strands shall be smooth, uniform and free from all imperfections such as spills and splits, die marks, scratches, abrasions and links after drawing and also after stranding.

7.2 The finished material shall have minimum brittleness as it will be subjected to appreciable vibration while in use.

7.3 The steel strands shall be hot dip galvanised and shall have a minimum zinc coating of 270 gms/sq.m of the uncoated wire surface. The zinc coating shall be smooth, continuous, of uniform thickness, free from imperfections and shall withstand four dips after stranding in standard Preece test. The steel wire rod shall be of such quality and purity that, when drawn to the size of the strands specified and coated with zinc, the finished strands shall be of uniform quality and have the same properties and characteristics.

7.4 The steel strands shall be preformed and post formed in order to prevent the spreading of strands in case stranded wire is cut. Care shall be taken to avoid damage to galvanisation during pre forming and post forming operations.

7.5 To avoid susceptibility towards wet sewerage strains (white rust), the finished material shall be provided with a protective coating of boiled linseed oil.

8.0 JOINTS IN WIRES

There shall be no joint of any kind in the finished steel wire strand entering into the manufacture of the stranded wire. There shall be no strand joints or strand splices in any length of the completed stranded steel wire.

9.0 STANDARD LENGTH

9.1 The standard lengths and weights in each drum coils shall be as mentioned below.

a) 7/3.15 mm. : For Earthwire - Standard length of 2000 mtrs. in each drum.

9.2 Supplier shall also indicate the maximum single length above the standard length, he can manufacture, in the guaranteed technical particulars. This is required for special stretches like river crossing etc. The Purchaser reserves the right to place orders for the above length to the extent of 5 percent of the total ordered quantity on the same terms and conditions applicable for the standard lengths during the pendency of the contract.

10.0 SCALE OF SAMPLING AND CRITERIA FOR CONFORMITY:

Samples from each lot shall be tested for ascertaining the conformity to the requirements of the groundwire specified herein. The no. of coils to be selected shall be as below:

No.of coils in the lot	**No. of coils to be selected.**
upto 25	3
26-50	5
51-150	8
151-300	10
301 and above	13

The coil selected shall be tested for length of the lay and joints. The lot shall be declared conforming to the requirements of these characteristics if all the coils are found satisfactory. One test speciman from each wire of the strand shall now be drawn from every selected coil and subjected to tensile test, ductility test and coating test. One test speciman of the completed strand from each coil shall be subjected to tensile strength. The lot shall be declared conforming to the requirement of these characteristics if all the test specimans satisfy the relevant requirements.

The guaranteed technical particulars for the materials being supplied shall be provided with the bid as specified in the Annexure-III enclosed. The bid received without these particulars shall be treated as non-responsive.

11.0 TESTS :

11.1 The following acceptance tests shall be conducted specified in IS:398 (Part-2)-1976 and IS:2141-1979.

a) <u>On individual wires from stranded wire:</u>

i) Breaking load and Elongation test
ii) Ductility test
iii) Wrapping test
iv) Galvanizing tests

b) <u>On stranded wire:</u>

i) Measurement of lay lengths

11.2 The following routine tests shall be conducted as specified in IS: 2141.

a) <u>On individual wires before stranding:</u>

i) Ductility test
ii) Measurement of diameter

b) <u>On individual wires after stranding:</u>

i) Tensile and elongation test
ii) Galvanizing tests

12.0 VERIFICATION OF LENGTH AND WEIGHT

Ten percent of the drums offered for inspection shall be checked for the length and weight of the wire declared by the supplier. One empty drum will be weighed and wire will be rewound on the empty drum and the wound drum will be reweighed. The length of the wire will be measured by using a certified measuring counter by appropriate weight and measurement authority. The cost towards unreeling of wire drums for verifying the wire length and weight shall be to the supplier's account.

13.0 THREE copies of manufacturer's test certificates shall be submitted by the supplier to the purchaser for approval immediately after such tests have been conducted on the G.S.Strands and the wire.

13.1 The purchaser reserves the right to inspect the material at manufacturer's works before despatch.

14.0 CHEMICAL ANALYSIS:

One sample shall be drawn from the lot for chemical analysis. The lot shall be declared in conformity with the requirements if all the test results satisfy relevant requirements.

15.0 INSPECTION:

15.1 The Purchaser’s representative shall, at all times, be entitled to have access to the works and all places of manufacture where the High Tensile Galvanised Stranded steel wire shall be manufactured and the representative shall have full facilities for unrestricted inspection of the Supplier's works, raw materials and process of manufacture and conducting necessary tests as detailed herein.

15.2 The Supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of the wire in its various stages so that arrangements can be made for inspection.

16.0 REJECTION AND RETEST

Should any one of the test pieces first selected fail to pass the tests, three further samples from the same batch shall be selected, one of which shall be from the length from which the original test sample was taken, unless that length has been withdrawn by the supplier.

16.1 Should all the three test pieces from these additional samples satisfy the requirements of the tests, the batch represented by these samples shall be deemed to comply with the standard. Should the test pieces from any of the three additional samples fail, the batch represented shall be deemed not to comply with the standard.

16.2 If the test pieces for tests on strands taken from finished wire on a drum selected for such tests fail in the retest, the manufacturer shall test each and every drum and submit them for re-inspection.

16.3 All rejected material shall be suitably marked and segregated.

16.4 No material shall be despatched from its point of manufacture before it has been satisfactorily inspected and tested, unless the inspection is waived off by the Purchaser in writing. In the later case also, the wire shall be despatched only after satisfactory testing for all tests specified herein has been completed and approved by the Purchaser.

16.5 The acceptance of any quantity of material shall in no way relieve the Supplier of any of his responsibilities for meeting all requirements of the specification, and shall not prevent subsequent rejection if such material is later found to be defective.

16.6 The Tenderer shall indicate the name(s) of reputed inspection agencies. The inspection charges will be borne by the Tenderer. However, the purchaser reserves the right to appoint at its cost any inspection agency to carry out the inspection & the expenditure incurred therein is to be borne by the Purchaser.

17.0 QUALITY ASSURANCE PLAN : Refer Clause-3 of Section-I, General

18.0 DOCUMENTATION: The Tenderer shall furnish along with the bid the dimensional drawings of the drum as per IS : 1778 / 1980 column no.2.

19.0 PACKING AND FORWARDING :

19.1 The earth wire portion of the 7/3.15 mm HTGS wire shall be supplied in non-returnable reels or drums of non-perishable or treated wood conforming to IS:9178-1961, IS:1778-1980 or its latest revision specification for reels and drums for bare wires BS-1659 or any other authoritative standard. Each drum shall be provided with a lable fixed firmly on it bearing the following information.

a) Trade name, if any
b) Name of manufacturer
c) Type of wire, size and length of wire
d) Net weight of the wire
e) Gross weight of the drum or reel
f) Name of the consigneee

The wire shall be packed in the drums in such way that the drums will withstand the streses and rough handling during transit, storing and stringing operations. The earth wire shall be protected from weather with water-proof paper before winding and after winding. The standard length of the earthwire shall be 2.00 KM for 7/3.15 mm in each drum as per clause 5.3.1 of this specification. There should not be more than one standard length in a drum. The tenderer should agree to supply the wire exceeding standard length in a drum for special river crossings etc., when specifically asked for. Each drum should be provided with a triangular iron coller on either side of the axle hold for each rolling and to protect the drum from damage during rolling/winding operation. Proper care should be taken to prevent the nails from damaging the steel wire.

20.0 ELECTRICAL SYSTEM PARTICULARS

		220kV	132 kV
a)	System Voltage(kV rms)	220	132
b)	Max Voltage (kV rms)	245	145
c)	Lighting impulse with stand voltage (Dry & Wet) (Kvp.)	1050	650

d)	Power frequency withstand voltage (Wet) (Kvp.)	460	280
e)	Short circuit level & duration	50 kA for 1 sec.	40kA for 1 sec.

TEST PROCEDURES

i)	Quality of Zinc	:	As per IS-209-1966 or its latest revision
ii)	Galvanising	:	As per IS-2629-1966 or its latest revision
iii)	Mass of Zinc coating	:	As per IS-6745-1972 or its latest revision
iv)	Uniformity of zinc coating	:	As per IS-2633-1972 or its latest revision
v)	Freedom from defects	:	As per IS-2629-1966 or its latest revision
vi)	Breaking Load Test		
vii)	Wrapping Test		
viii)	Measurement of lay lengths		

ix) DUCTILITY TEST:

The wire shall be subjected to wrapping test in accordance with the latest IS:1755 or any other authoritative standard. When wrapped eight times around its own diameter and on being subsequently straightened the wire shall not break or split.

xi) TENSILE AND ELONGATION TEST:

The ends of the strand shall be suitably prepared by either filling in the interstices with finer wire or other suitable means, to minimize failure at the testing machine grips. The load shall be applied at a steady rate not exceeding 100 N/Sq.mm per second. An initial load of 10% of the specified breaking load shall be applied to the test piece, at which stage the distance apart of the testing machine grips shall be 30 times the dia. of the strand subject to a minumum of 100 mm.

The elongation shall be determined as the % increase in separation between and the grips from the position after application of the initial load to the position at initial failure in the test piece. Elongation tests shall be made on lengths of strand, which do not contain wire joints. The percentage elongation of the strand (before stranding) shall not be less than 4%. The minimum-breaking load (or the minimum ultimate tensile strength) of the strand shall be 56.98 kN. The tensile strength of the finished strand (stranded wire) shall not be less than 95% of the aggregate of single wires.

xii) (a) ON INDIVIDUAL WIRES BEFORE MANUFACTURE :

i) Ductility Test

ii) Measurement of Diameter

(b) ON INDIVIDUAL WIRES AFTER STRANDING :

i) Tensile and elongation test
ii) Galvanising tests

xiii) COATING TEST:

The uniformity of zinc coating shall be tested as per IS:2633 or any other authoritative standard. The zinc shall conform to IS:209-1979. Hot dip galvanising of wire shall conform to IS:2629-1966. The zinc coating shall conform to heavy coating as laid down in IS:4826-1968 (Galvanised coating on round steel wires).

21.0 QUANTITY AND DELIVERY REQUIREMENTS : Refer Clause-5 of Section-I, General

22 SUPERVISION SERVICES : Refer Clause-6 of Section-I, General

GUARANTEED TECHNICAL PARTICULARS OF THE 7/3.15 mm
STRANDED HIGH TENSILE GALVANISED STEEL WIRE

(to be filled and furnished along with the Bid (part-1)

1.	Maker's Name, address and Country.	:	
2.	Percentage of carbon content of the steel wire	:	
3.	Particulars of Single wire before stranding	:	
	a) No.of strands	:	
	b) Diameter	:	mm
	c) Standard Sectional area	:	Sq.mm
	d) Minimum ultimate tensile strength	:	KN
	e) Minimum breaking load	:	KN
	f) Final stress in steel wires	:	KN
	g) Minimum elongation (Percentage)	:	
4.	a) Uniformity of coating of (number and duration of each dip)	:	
	b) Minimum weight of coating	:	gm/sq.m.
	c) Quality of zinc	:	
	d) Process of galvanizing	:	
5.	Standard overall diameter of stranded wire	:	Mm
6.	Area of cross section of stranded wire	:	Sq.mm
7.	Direction of lay of outer layer	:	
8.	Guaranteed ultimate tensile strength of wire	:	KN
9.	Maximum working tension	:	KN
10.	Resistance in Ohms per Km at 20 Deg C of stranded wire	:	Ohms/Km.
11.	Tolerance if any of standard length.(Percent of standard length)	:	
12.	Co-Efficient of linear expansion	:	Per Deg.C
13.	Modulus of elasticity of wire	:	Kg/sq.m
14.	Weight of wire	:	Kg/Km
15.	Tensile strength of joints expressed as a percentage of the strength of wire	:	
16.	a) Standard length of wire in each drum for 7/3.15 mm size	:	Km
17.	Maximum single length which can be manufactured	:	Km
18.	Standard weight of HTGS wire in each drum	:	Kg
19.	Weight of drum in Kg with lagging	:	Kg
20.	Gross weight of the drum	:	Kg
21.	Dimension of the drum in Cms.	:	Cm
22.	Standard according to which the earth wire will be manufactured.	:	
23.	Standard according to which the drum is manufactured	:	
24.	Other particulars, if any	:	

SIGNATURE OF THE BIDDER
WITH DATE & SEAL